
You are here: [Home](#) / [Electronics](#) / What is Open Drain : Configuration & Its Applications

What is Open Drain : Configuration & Its Applications

December 7, 2021 By WatElectronics

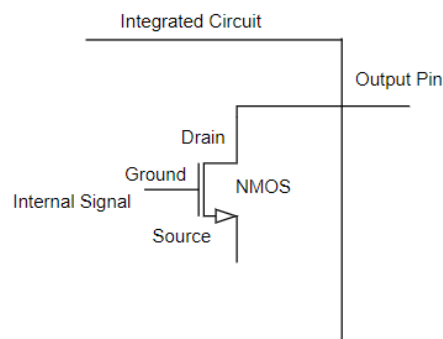
In order to have interference with the outside world using microcontrollers, it can be done through pins. These pins act as physical points for an integrated circuit where a PCB connection can be established. Each pin has its own circuitry, and it is configured to pins to allow interference with various kinds of circuits. And the most important point is to select the appropriate pin configuration in the design of an embedded system. Today, this article explains open-drain output configuration, how it is configured, its applications, and its advantages.

What is an Open Drain?



Circuit and Working

In most of the integrated circuits, the observed output configuration type is open-drain. As a replacement for outputting a signal with a corresponding current or voltage, the output is fed to the base terminal of the NPN **transistor**, and the collector terminal is externally connected to the integrated circuit pin. The emitter terminal has an internal connection with the grounded pin. When the output circuit is in MOSFET configuration, then it is termed as **open drain circuit** and it operates in the same way.



©WatElectronics.com

Open Drain Circuit



[HOME](#)[ARTICLES](#)[BASICS](#)[COMPONENTS](#)[PROJECTS](#)[COMMUNICATIONS](#)[MCQ](#)

device connected on the wire flows enough amount of current in order to pull the line to low and to the signal the start of a communication.

When all the devices which are connected to the signal wire are in a non-active state, then the pull-up makes the wire at a high voltage phase. When one or more devices function in the active state, then the voltage of the signal wire moves to low.

In the **open drain working**, the signal wire also functions in bi-directional which corresponds that the device can have both input/output signals on the wire at the same period of time. Along with the regulation of the pins state, which is connected to the signal wire, it can also recognize the level of voltage of the wire.

Even though the output of the open collector device is in the high state, the signal wire which is attached to it might be in low state because of the other device that is connected to the signal wire. So, with the bi-directional behavior, the **open collector** device can be useful for the interconnection of multiple devices on a single line.

Open Drain Configuration

The open-drain configuration is employed at the controller's output port in order to lessen the output

Download Manuals (PDF)

View, Download & Print User Manuals for Free!



In order to employ analog input pin as high-impedance input and a comparator is enabled state, then that pin has to be in the configuration of open collector and the related register bit has to be in '1' state.

In both of the scenarios, the point to be noted is that the open drain/collector is a part of the output transistor which is directly connected out to a pin that is on the external side of the integrated circuit.

The below picture represents the **open drain configuration**.

Types of Output Configurations in Comparison with Push-Pull and Pull Up Mode

In digital sensing applications, the crucial part is to select the suitable output configuration where the main options are open collector, push-pull and push-up. To know these configurations, let us know a few terms which are explained below:

Download Manuals (

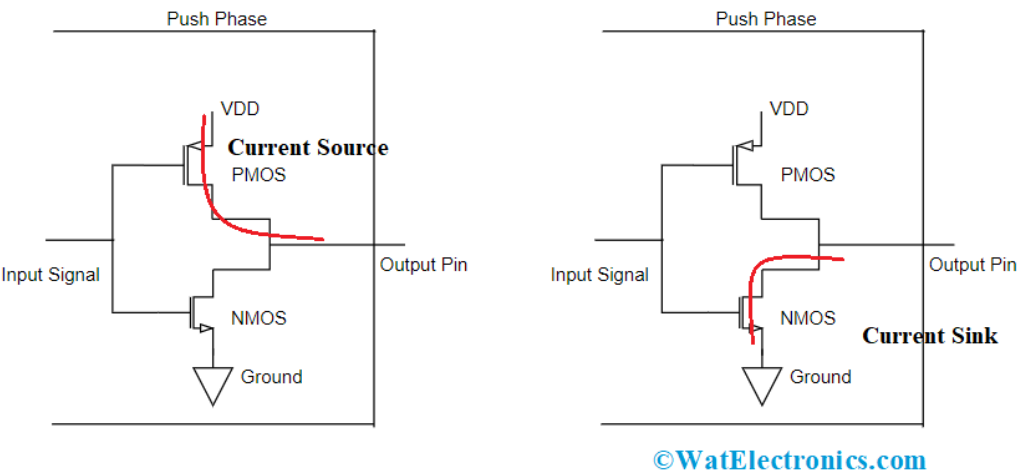


MCQ

to which it is connected. It is usually not driven actively and is in a floating state except it is connected to any other external device.

Open Drain v/s Push Pull

Push-pull output configuration is the most implemented type where the name itself indicates that this configuration holds the ability to drive two different output levels. One is to sink current from the load section and the other is to source current to the load section. This configuration is implemented through switches. The two scenarios push and pull representations are shown below:



Push Pull Output

The **open drain advantages and disadvantages** in comparison with push-pull configuration are:

▼

Download Manuals (

[HOME](#)[ARTICLES](#)[BASICS](#)[COMPONENTS](#)[PROJECTS](#)[COMMUNICATIONS](#)[MCQ](#)**Benefits**

- It can directly drive loads those are $\leq 10 \text{ mA}$
- It has high functional speeds in the range of 10KHz
- It has stable high and low outputs

Drawbacks

- It requires high power to operate when compared with push-pull because of drain via load resistor when the device is in ON condition
- It does not hold the ability to drive external loads
- The switching speed is minimal. It achieves faster switching speeds when minimal load resistor values are used
- It cannot combine output signals to multiple sensors to a single bus

Output

When the output is at high, then the pin will be connected to the power supply using NPN switch, and when the output low state, the pin gets connected to the ground using PNP switch

When the output is at high, then the pin will be connected to the ground using a switch, and when the output low state, the pin will be in a floating state as the switch is turned OFF

On the whole, push-pull configuration is selected when there is the requirement of minimal circuitry, and an open collector configuration is chosen for the requirements of interfacing with various system voltages, load switching, and when multiple device outputs are to be connected to a single bus.

Open Drain v/s Pull Up

[HOME](#)[ARTICLES](#)[BASICS](#)[COMPONENTS](#)[PROJECTS](#)[COMMUNICATIONS](#)[MCQ](#)

pull-up **resistors** are used. Upon the activation of the pull-up resistor, the input pin changes its state to V_{dd} .

Output mode in open-drain configuration simply does not have a PMOS transistor. When the switch is in the OFF state, the drain gets opened and output moves into a floating state. In this configuration, the pin is only pulled up but not pulled down. The open-drain configuration is of no use when it does not support pull-up functionality.

In order to make this configuration applicable in real-world scenarios, either an external or internal pull-up resistor has to be used. These days, all the **microcontrollers** are provided with a pull-up resistor for the output pin.

Applications

The applications of open drain include the following.

- The crucial application is in summing, weighting, and limiting both analog and digital logic.
- Used in vacuum fluorescent displays and in Nixie tubes
- Used in SCS-1 devices for the purpose of electrical signaling.

What is open-drain used for?

Open collector/drain presents high impedance either high or low voltage is applied to the gate terminal.

What is open drain and wired or?

The open-drain technique is used to establish communication between multiple devices bi-directionally on a single wire.

Is open-drain high impedance?

[HOME](#)[ARTICLES](#)[BASICS](#)[COMPONENTS](#)[PROJECTS](#)[COMMUNICATIONS](#)[MCQ](#)

open-drain configuration requires a pull-up resistor so that the output will be properly at high.

What is an open-drain buffer?

The open collector/drain buffer is 74LVC1G07 IC with open-drain output. This buffer when connected with other open-collector outputs is suitable for the application of active low (wired-OR) or wired-AND functions.

Please refer to this link to know more about [Analog Electronics MCQs](#).

So, this is all about the concept of the open drive. This article has clearly focused on explaining what is open-drain configuration, its comparison with [push-pull](#) and pull-up configurations, circuits, and applications. Also, know [how an open-drain output configuration is used in driving LEDs?](#)



[HOME](#)[ARTICLES](#)[BASICS](#)[COMPONENTS](#)[PROJECTS](#)[COMMUNICATIONS](#)[MCQ](#)

Collision Sensor : PinOut, Features, Specifications, Interfacing, Working & Its Applications

TM4C123 IC : PinOut, Features, Specifications, Interfacing & Its Applications

ISD1820 Voice Recording Module : PinOut, Features, Specifications, Interfacing & Its Applications

ZN414 IC : PinOut, Features, Specifications, Circuit, Working, Equivalent, Datasheet & Its Applications

XC6206 IC : PinOut, Features, Specifications, Circuit, Working, Datasheet & Its Applications

4N35 IC : PinOut, Features, Specifications, Vs 4N25, Interfacing & Its Applications

AZ1084 IC : PinOut, Features, Specifications, Circuit, Working, Datasheet & Its Applications

HK620 IC : PinOut, Features, Specifications, Circuit, Working, Datasheet & Its Applications

DS18B20 Temperature Sensor : PinOut, Features, Specifications, Interfacing & Its Applications

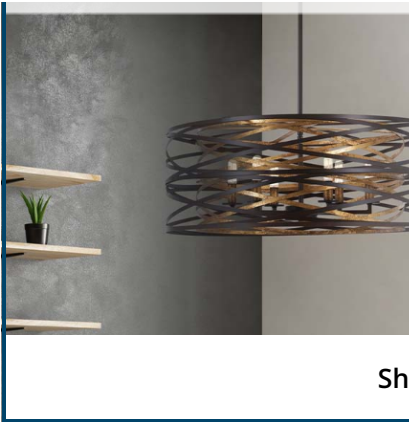
MIC5219 IC : PinOut, Features,

Specifications, Circuit, Working & Its Applications



[HOME](#) [ARTICLES](#) [BASICS](#) [COMPONENTS](#) [PROJECTS](#) [COMMUNICATIONS](#)

[MCQ](#)



CATEGORIES

Articles (20)

Basics (109)

Communications (54)

Components (206)

Digital Electronics (43)

Digital Signalling (3)

Electronics (225)

Embedded Systems (12)

Magnetism (5)

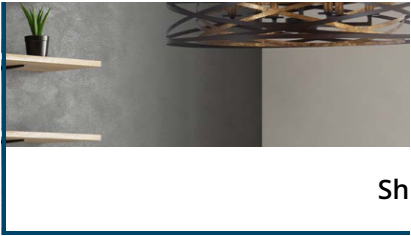
Microprocessors (3)

Modulation (1)



HOME ARTICLES BASICS COMPONENTS PROJECTS COMMUNICATIONS

MCQ



SUBSCRIBE TO OUR NEWSLETTER

Don't miss these articles!

Email Address *

[SUBSCRIBE TO NEWSLETTER](#)

We don't spam! Read our privacy policy for more info.



CATEGORY

MCQ

